QGIS Tutorial – Session 1

Introduction to Spatial Data for Spatial Epidemiology
Queensland, Australia
Duration: approximately 2 hours

Goal of this tutorial

In this first practical session, you will build a simple spatial epidemiology project in QGIS using Queensland data. You will work with the four spatial data types that will appear repeatedly throughout the course: polygons, points, lines and rasters.

The tutorial is designed for students who may have little or no previous experience with QGIS. Follow the instructions in order. The purpose is not simply to learn where QGIS buttons are located, but to understand how different types of geographic information can be combined to answer public health questions.

The public health question

Throughout the tutorial, keep one broad question in mind:

Question: Where is the simulated disease burden concentrated in Queensland, and how can population distribution, socioeconomic conditions, remoteness, rainfall, health facilities and roads help us understand the geographic context?

We will not try to answer this question statistically in Session 1. Instead, we will learn how to organise, explore and visualise the spatial information that would be needed for a spatial epidemiological analysis.

Before you begin

Create a course folder on your computer. Keep all data and your QGIS project inside this folder. A simple folder structure is:

- Spatial_Epidemiology_Course
- Spatial_Epidemiology_Course/Data
- Spatial_Epidemiology_Course/Projects
- Spatial_Epidemiology_Course/Outputs

Good GIS habit: Do not move or rename your data files after adding them to a QGIS project. A QGIS project usually stores links to the source data rather than copying the data into the project file.

Data used in this tutorial

Your course folder should contain the following datasets. The exact filenames may vary slightly depending on the final course package.

Dataset	GIS type	What it represents	Why we need it
Queensland SA2 course GeoPackage	Polygon	SA2 statistical areas with simulated health outcome and covariates	Main epidemiological analysis layer
Health facilities	Point	Locations of health services	Introduces point data and healthcare accessibility
Roads	Line	Queensland road	Introduces line data and

		network	movement/access
Population raster	Raster	Gridded population distribution	Shows population within, not only between, SA2s

Learning outcomes

By the end of this tutorial, you should be able to:

- Create and save a QGIS project.
- Recognise point, line, polygon and raster spatial data.
- Add vector and raster layers to QGIS.
- Navigate around a map using zoom and pan tools.
- Use the Layers panel and control layer order and visibility.
- Use Identify Features and open an attribute table.
- Understand the basic purpose of a Coordinate Reference System (CRS).
- Create a choropleth map of a health outcome.
- Map continuous and categorical explanatory variables.
- Explore simple spatial relationships between SA2s and health facilities.
- Understand why population rasters can be useful in spatial epidemiology.
- Create a simple map layout for communication.

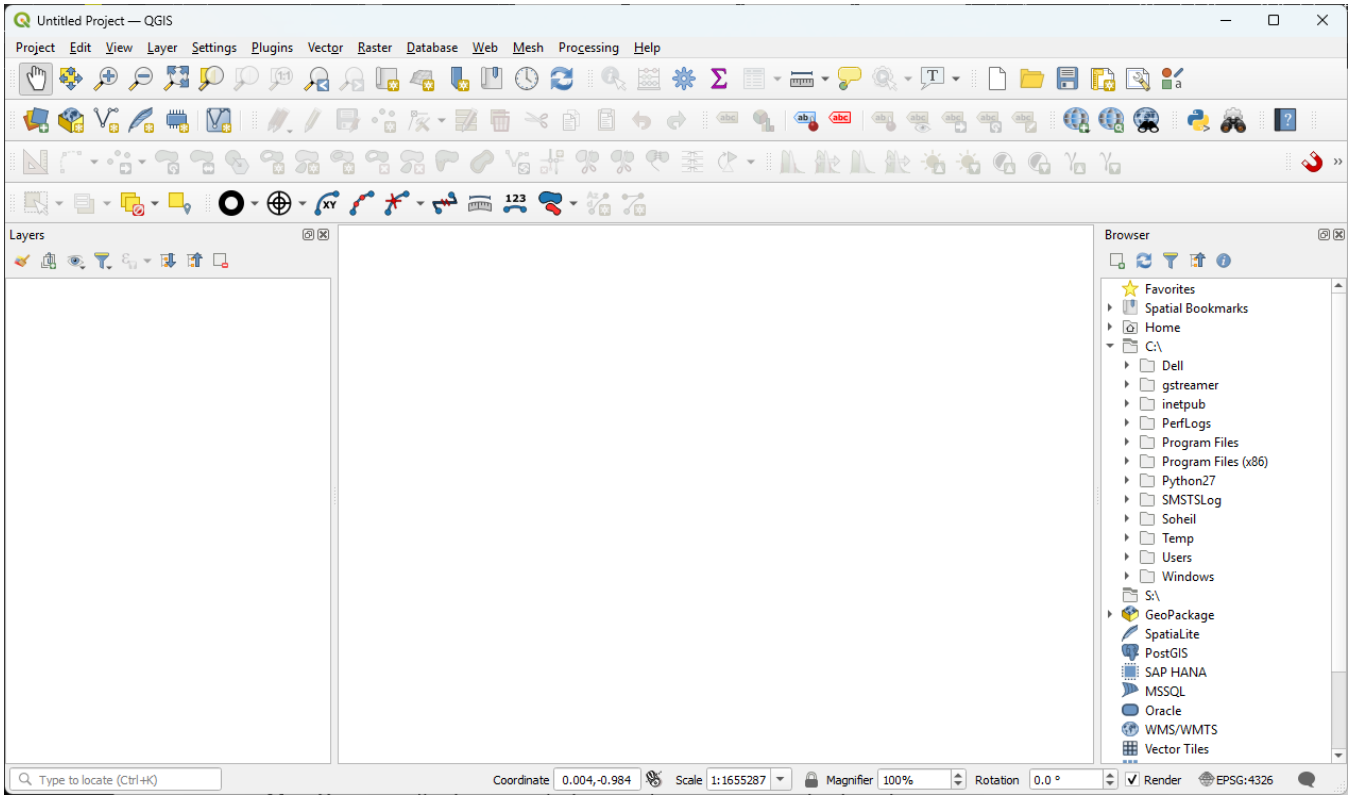
Part 1. Open QGIS and create your project

1.1 Open QGIS

Step 1. Open QGIS from your computer.

Step 2. Wait until the main QGIS window appears.

The exact appearance may differ slightly depending on your QGIS version and operating system.



The empty QGIS interface with the Map Canvas, Layers panel, Browser panel and main toolbars visible.

1.2 Become familiar with the QGIS interface

Before adding any data, locate the following parts of the QGIS window:

- Map Canvas – the large central area where maps are displayed.
- Layers panel – usually on the left; shows all layers in the current project.
- Browser panel – allows you to browse folders, GeoPackages and other data sources.
- Menu bar – contains menus such as Project, Layer, Vector, Raster and Processing.
- Toolbars – contain shortcuts for common tools such as zoom, pan, identify and selection.
- Status bar – at the bottom; shows information including coordinates, scale and project CRS.

Key concept: QGIS is not the data itself. It is software that allows you to view, manage, analyse and map spatial data.

1.3 Save the project

Step 1. Click Project > Save As.

Step 2. Navigate to your Projects folder.

Step 3. Enter the filename: Session1_Queensland_Spatial_Epidemiology.qgz

Step 4. Click Save.

Expected result: A .qgz QGIS project file should now exist in your Projects folder.

Common mistake: Saving the project does not automatically copy all your spatial datasets into the project. Keep the Data folder with the project.

Part 2. Understand the spatial data types before loading them

2.1 Vector data

Vector data represent discrete geographic features. There are three basic vector geometries.

Point data represent a specific location. In this tutorial, health facilities are points.

Line data represent linear features. In this tutorial, roads are lines.

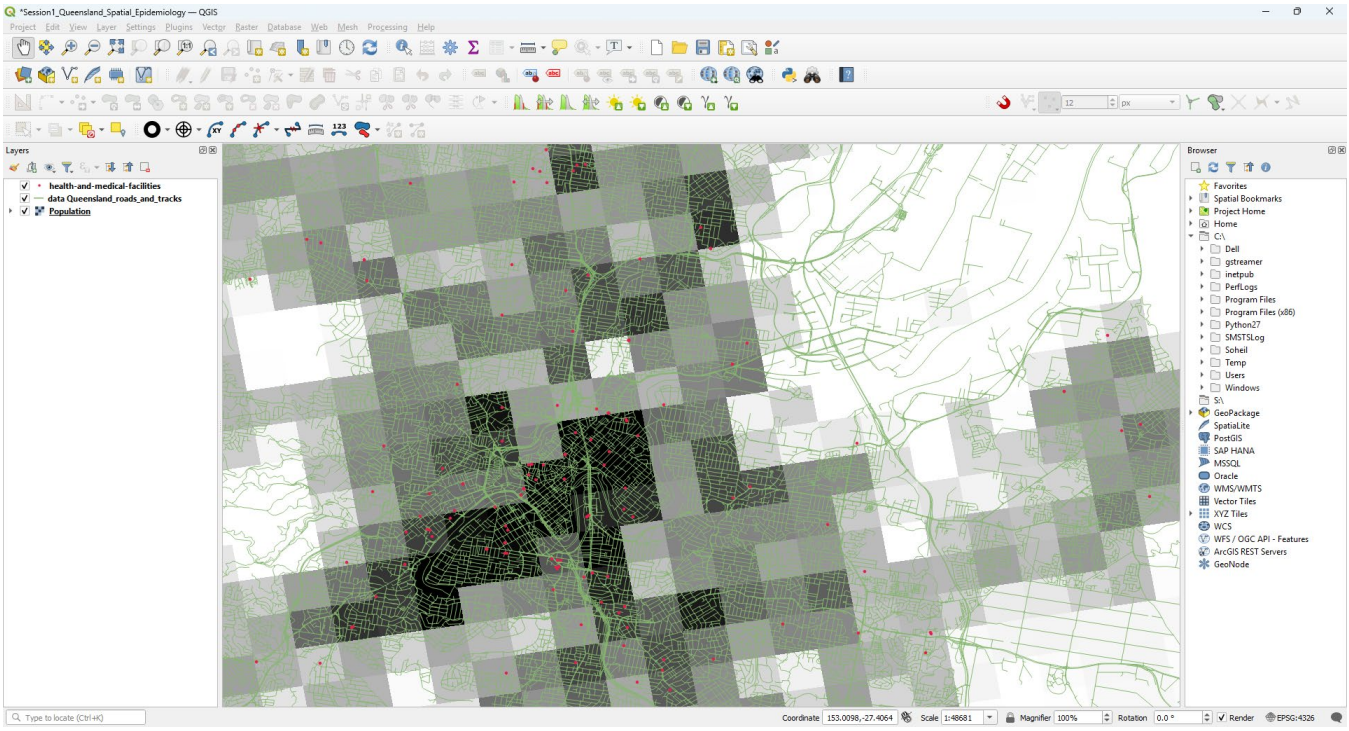
Polygon data represent areas with boundaries. In this tutorial, SA2 statistical areas are polygons.

Think: If you wanted to map hospitals, rivers, local government areas and individual households, which geometry would you use for each?

2.2 Raster data

Raster data represent geographic space as a regular grid of cells or pixels. Every cell stores a value. Our population raster stores information about population distribution across space.

Key concept: Vector data describe geographic objects. Raster data describe geographic space as cells.



Line (roads), points (healthcare facilities) and raster (population) layers in QGIS

Part 3. Add the Queensland SA2 polygon layer

3.1 Add the GeoPackage

Step 1. Click Layer > Add Layer > Add Vector Layer.

Step 2. Under Source type, select File.

Step 3. Click the three dots (...) beside Vector Dataset(s).

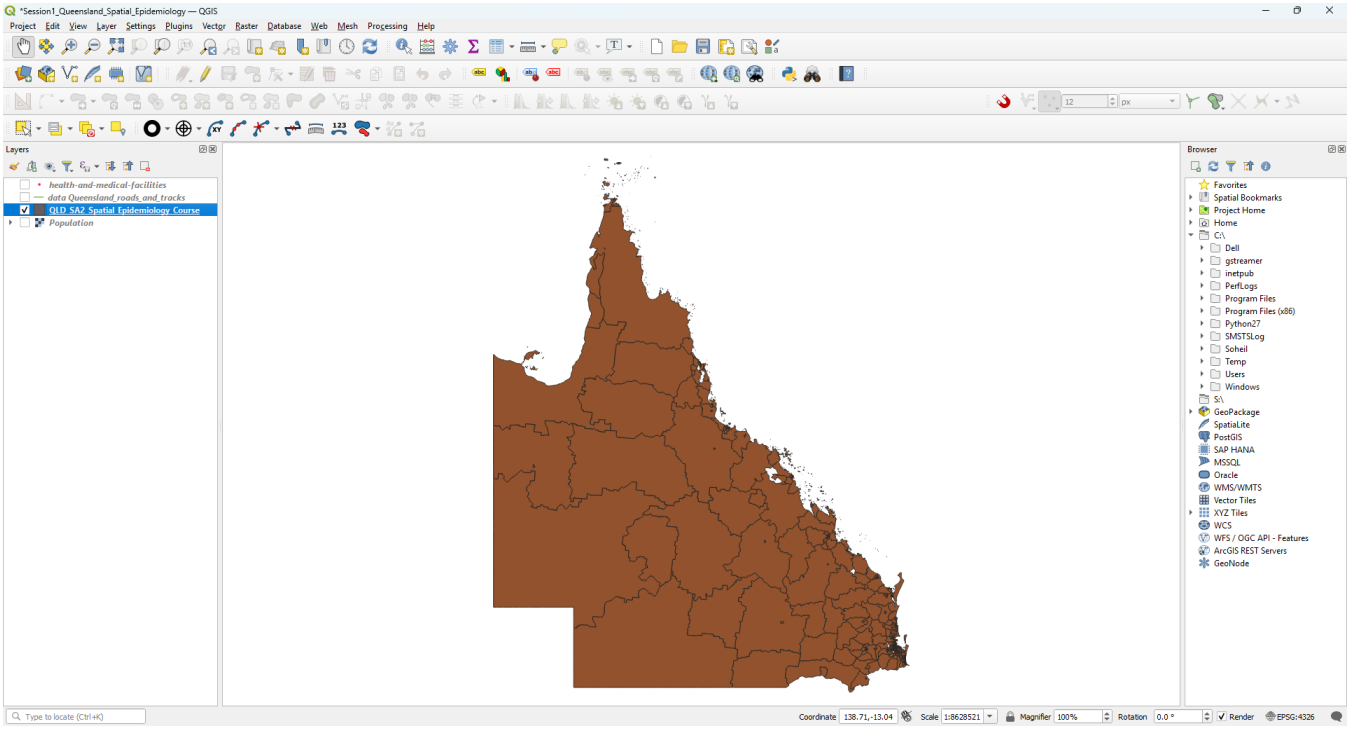
Step 4. Navigate to the Data folder and select the Queensland SA2 course GeoPackage.

Step 5. Click Add.

Step 6. If QGIS asks you to choose a layer from the GeoPackage, select the final Queensland SA2 course layer.

Step 7. Click Add, then Close.

Expected result: Queensland should appear in the Map Canvas as a set of SA2 polygons. The layer should also appear in the Layers panel.



QGIS after adding the Queensland SA2 layer, showing the whole of Queensland and the layer name in the Layers panel.

3.2 Zoom to the layer

Step 1. In the Layers panel, right-click the Queensland SA2 layer.

Step 2. Click Zoom to Layer.

QGIS should adjust the map extent so the entire layer is visible.

Tip: If you become lost while zooming around the map, Zoom to Layer is a quick way to return to the full dataset.

3.3 Navigate around the map

Step 1. Use the mouse wheel to zoom in and out.

Step 2. Select the Pan Map tool, usually shown as a hand icon.

Step 3. Click and drag the map to move to another part of Queensland.

Step 4. Zoom into South East Queensland and observe how SA2 polygons become easier to distinguish.

Pause and interpret: Why are some SA2 polygons geographically very large while others are very small? Think about the relationship between administrative geography and population density.

Part 4. Connect the map to the data

4.1 Use Identify Features

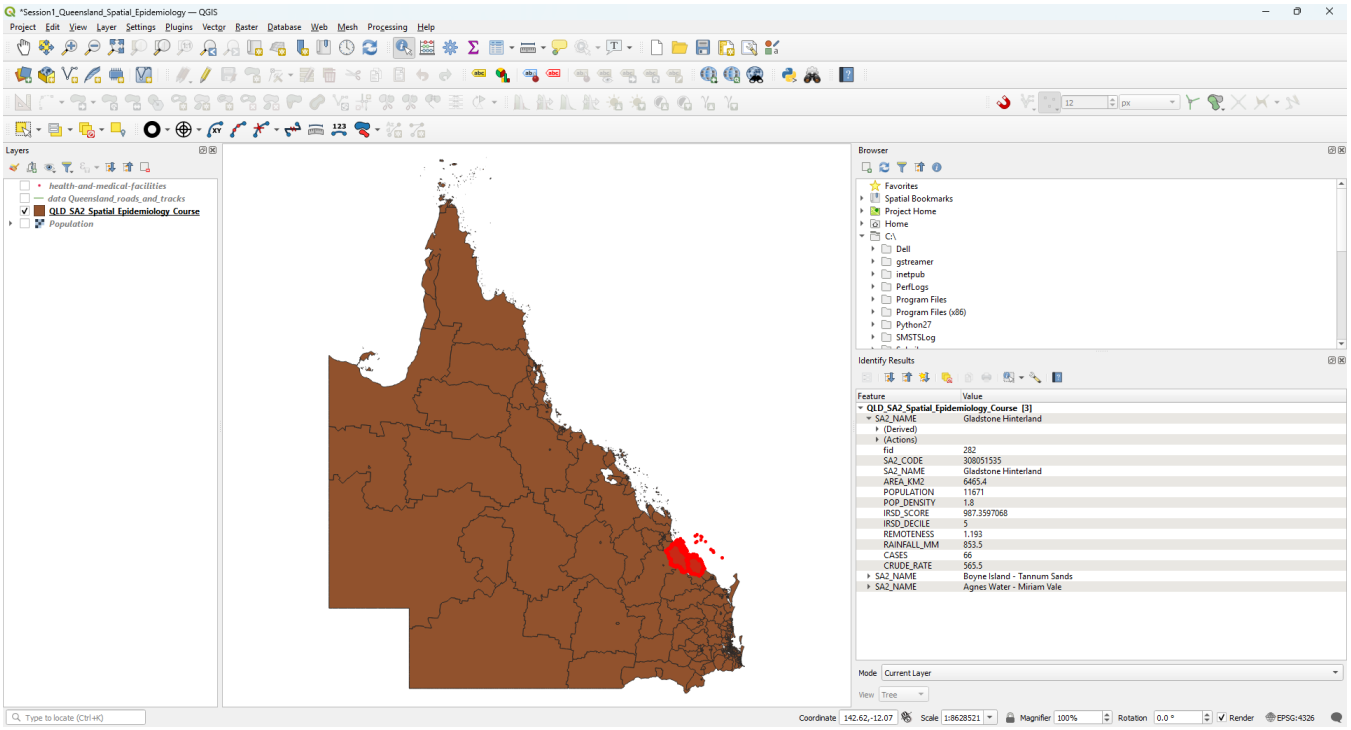
Step 1. Click the Identify Features tool on the toolbar.

Step 2. Click one SA2 polygon.

Step 3. Look at the Identify Results panel.

The information shown belongs to the SA2 that you clicked. You should see identifiers and several variables prepared for the course.

Expected result: One geographic polygon on the map is linked to a set of attribute values.



Identify Features results for one SA2, with the selected SA2 visible on the map and its attributes visible in the Identify Results panel.

Key concept: A GIS layer contains both geometry and attributes. Geometry tells us where a feature is. Attributes tell us what we know about it.

4.2 Open the attribute table

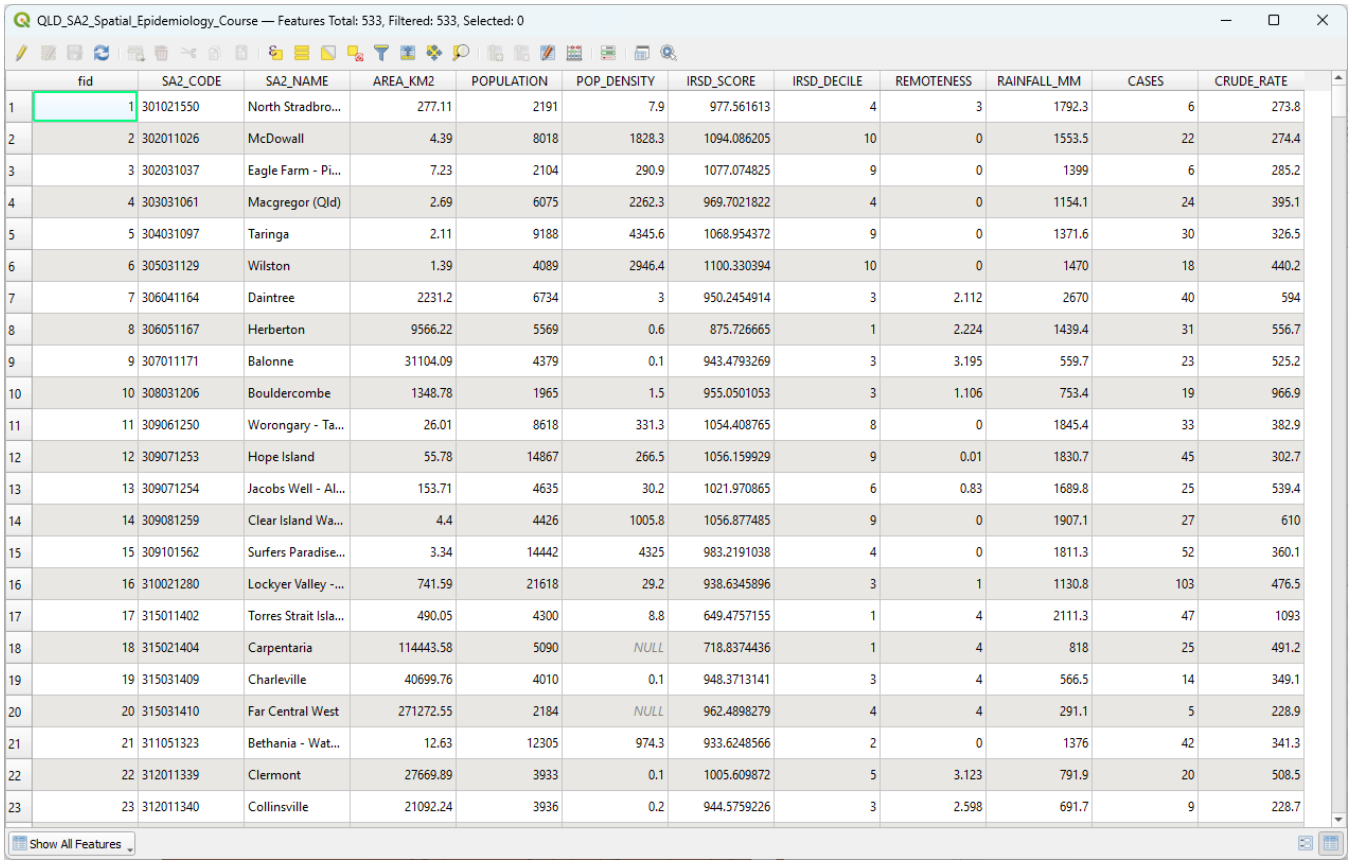
Step 1. Right-click the Queensland SA2 layer in the Layers panel.

Step 2. Click Open Attribute Table.

Step 3. Scroll horizontally through the columns.

Step 4. Look for fields representing SA2 code/name, population, the simulated health outcome, SEIFA/IRSAD, remoteness and rainfall.

Each row represents one SA2. Each column represents one characteristic of that SA2.



	fid	SA2_CODE	SA2_NAME	AREA_KM2	POPULATION	POP_DENSITY	IRSD_SCORE	IRSD_DECILE	REMOTENESS	RAINFALL_MM	CASES	CRUDE_RATE
1	1	301021550	North Stradbro...	277.11	2191	7.9	977.561613	4	3	1792.3	6	273.8
2	2	302011026	McDowall	4.39	8018	1828.3	1094.086205	10	0	1553.5	22	274.4
3	3	302031037	Eagle Farm - Pi...	7.23	2104	290.9	1077.074825	9	0	1399	6	285.2
4	4	303031061	Macgregor (Qld)	2.69	6075	2262.3	969.7021822	4	0	1154.1	24	395.1
5	5	304031097	Taringa	2.11	9188	4345.6	1068.954372	9	0	1371.6	30	326.5
6	6	305031129	Wilston	1.39	4089	2946.4	1100.330394	10	0	1470	18	440.2
7	7	306041164	Daintree	2231.2	6734	3	950.2454914	3	2.112	2670	40	594
8	8	306051167	Herberton	9566.22	5569	0.6	875.726665	1	2.224	1439.4	31	556.7
9	9	307011171	Balonne	31104.09	4379	0.1	943.4793269	3	3.195	559.7	23	525.2
10	10	308031206	Bouldercombe	1348.78	1965	1.5	955.0501053	3	1.106	753.4	19	966.9
11	11	309061250	Worongary - Ta...	26.01	8618	331.3	1054.408765	8	0	1845.4	33	382.9
12	12	309071253	Hope Island	55.78	14867	266.5	1056.159929	9	0.01	1830.7	45	302.7
13	13	309071254	Jacobs Well - Al...	153.71	4635	30.2	1021.970865	6	0.83	1689.8	25	539.4
14	14	309081259	Clear Island Wa...	4.4	4426	1005.8	1056.877485	9	0	1907.1	27	610
15	15	309101562	Surfers Paradise...	3.34	14442	4325	983.2191038	4	0	1811.3	52	360.1
16	16	310021280	Lockyer Valley -...	741.59	21618	29.2	938.6345896	3	1	1130.8	103	476.5
17	17	315011402	Torres Strait Isla...	490.05	4300	8.8	649.4757155	1	4	2111.3	47	1093
18	18	315021404	Carpentaria	114443.58	5090	NULL	718.8374436	1	4	818	25	491.2
19	19	315031409	Charleville	40699.76	4010	0.1	948.3713141	3	4	566.5	14	349.1
20	20	315031410	Far Central West	271272.55	2184	NULL	962.4898279	4	4	291.1	5	228.9
21	21	311051323	Bethania - Wat...	12.63	12305	974.3	933.6248566	2	0	1376	42	341.3
22	22	312011339	Clermont	27669.89	3933	0.1	1005.609872	5	3.123	791.9	20	508.5
23	23	312011340	Collinsville	21092.24	3936	0.2	944.5759226	3	2.598	691.7	9	228.7

The SA2 attribute table showing several rows and the main course variables.

4.3 Link a row to a place

Step 1. Click the row number of one SA2 in the attribute table.

Step 2. The corresponding polygon should become selected on the map.

Step 3. Close the attribute table.

Step 4. Use the selection tool on the map to select a different SA2.

Step 5. Open the attribute table again and display selected features.

Expected result: You should be able to move in both directions: from a place on the map to its data, and from a row in the table to its geographic location.

Part 5. A first look at Coordinate Reference Systems

5.1 Find the layer CRS

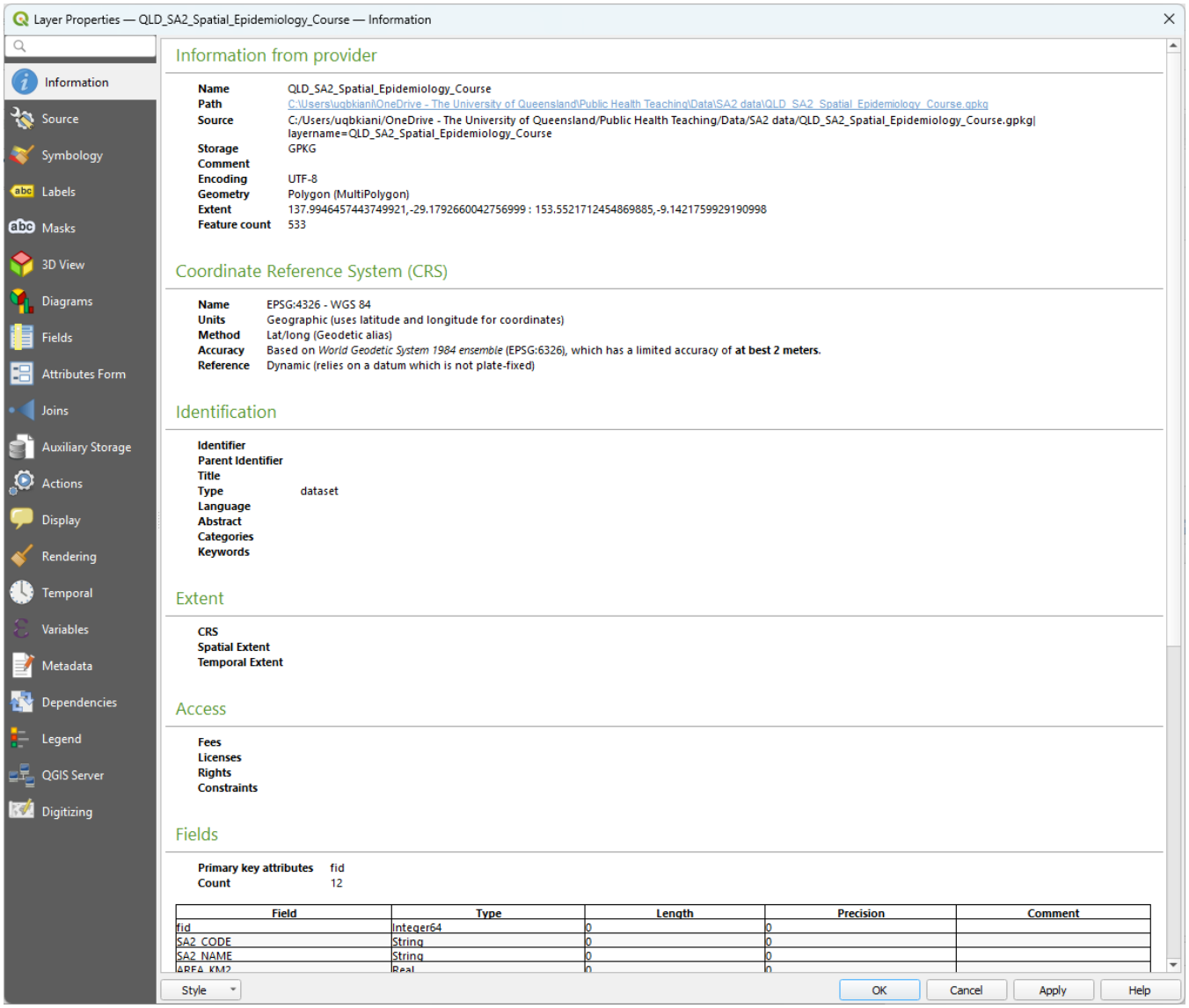
Step 1. Right-click the Queensland SA2 layer.

Step 2. Choose Properties.

Step 3. Open the Information or Source section.

Step 4. Find the Coordinate Reference System (CRS).

Step 5. Click Cancel or OK to close Layer Properties.



Layer Properties showing the CRS information for the Queensland SA2 layer.

5.2 Find the project CRS

Step 1. Look at the bottom-right corner of the QGIS window.

Step 2. Find the CRS indicator for the project.

A CRS defines how geographic positions are represented. Geographic coordinate systems commonly use longitude and latitude. Projected coordinate systems represent the curved Earth on a flat surface and commonly use metres.

Key concept: The Project CRS controls how QGIS displays the project. A layer also has its own CRS describing how its coordinates are stored.

Why are we doing this?: Later in spatial epidemiology we will calculate distances, areas, buffers and accessibility. Those operations require careful attention to CRS.

Common mistake: Changing the Project CRS is not the same as permanently reprojecting a layer.

Part 6. Add health facilities – point data

6.1 Add the health facility layer

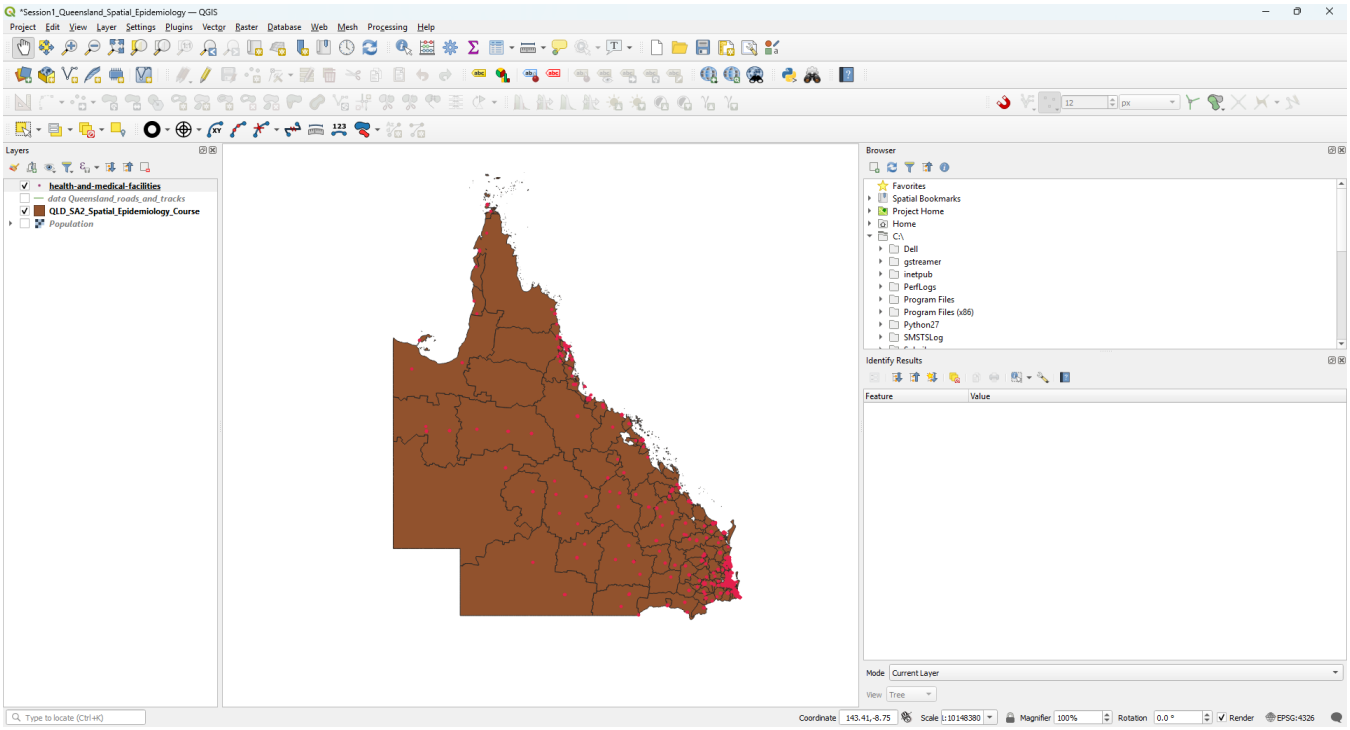
Step 1. Click Layer > Add Layer > Add Vector Layer.

Step 2. Select File as the source type.

Step 3. Browse to the health facilities dataset in the Data folder.

Step 4. Click Add, then Close.

Expected result: Health facilities should appear as points on top of the SA2 polygons.



Queensland SA2 polygons with health facility points displayed and both layers visible in the Layers panel.

6.2 Make the facilities easier to see

Step 1. Right-click the health facilities layer.

Step 2. Choose Properties > Symbology.

Step 3. Click the marker symbol.

Step 4. Increase the marker size until the facilities are clearly visible.

Step 5. Click Apply, then OK.

Key concept: Changing symbology changes only how the data look. It does not change the underlying locations or attributes.

6.3 Explore health facility attributes

Step 1. Use Identify Features and click a health facility.

Step 2. Review the available attributes.

Step 3. Open the health facilities attribute table.

Step 4. Inspect several rows.

Pause and interpret: Where do facilities appear concentrated? Are there large areas with relatively few facilities? What additional information would you need before concluding that healthcare access is poor?

Part 7. Add roads – line data

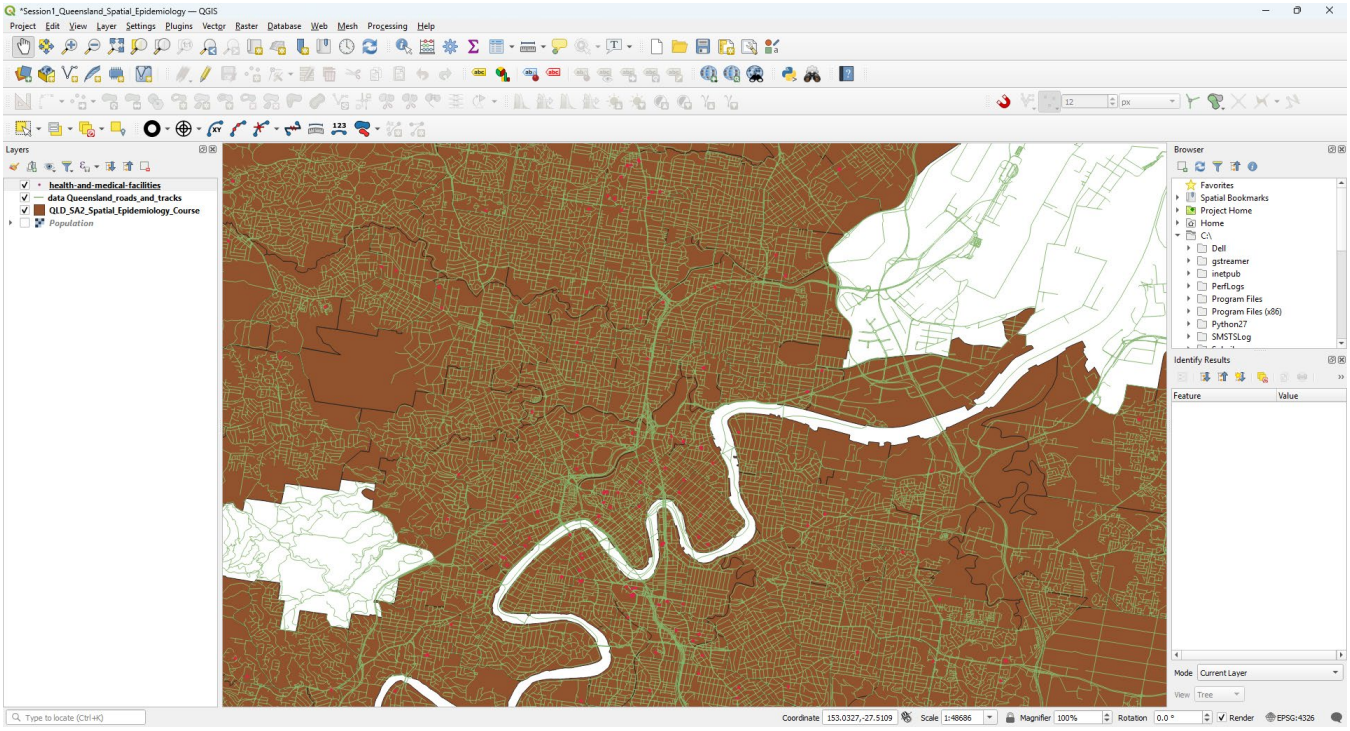
7.1 Add the road layer

Step 1. Click Layer > Add Layer > Add Vector Layer.

Step 2. Browse to the Queensland roads dataset.

Step 3. Click Add, then Close.

Expected result: Roads should appear as line features across Queensland.



A zoomed view showing SA2 polygons, roads and health facility points together.

7.2 Understand layer order

Look at the Layers panel. QGIS draws layers from the bottom upward. A layer higher in the list can cover a layer below it.

Step 1. Drag the health facilities layer near the top.

Step 2. Place the roads layer below health facilities.

Step 3. Place the SA2 layer below the roads.

Step 4. Turn individual layers off and on using their checkboxes.

Good GIS habit: Arrange your layers so the most important point and line information is not hidden by polygon or raster layers.

7.3 Explore roads and facilities together

Step 1. Zoom into Brisbane or another populated area.

Step 2. Observe the relationship between health facilities and the road network.

Step 3. Pan toward a more remote area and compare the pattern.

Think: Why might road-network travel time be more meaningful than straight-line distance when measuring access to a hospital?

Part 8. Add population – raster data

8.1 Add the population raster

Step 1. Click Layer > Add Layer > Add Raster Layer.

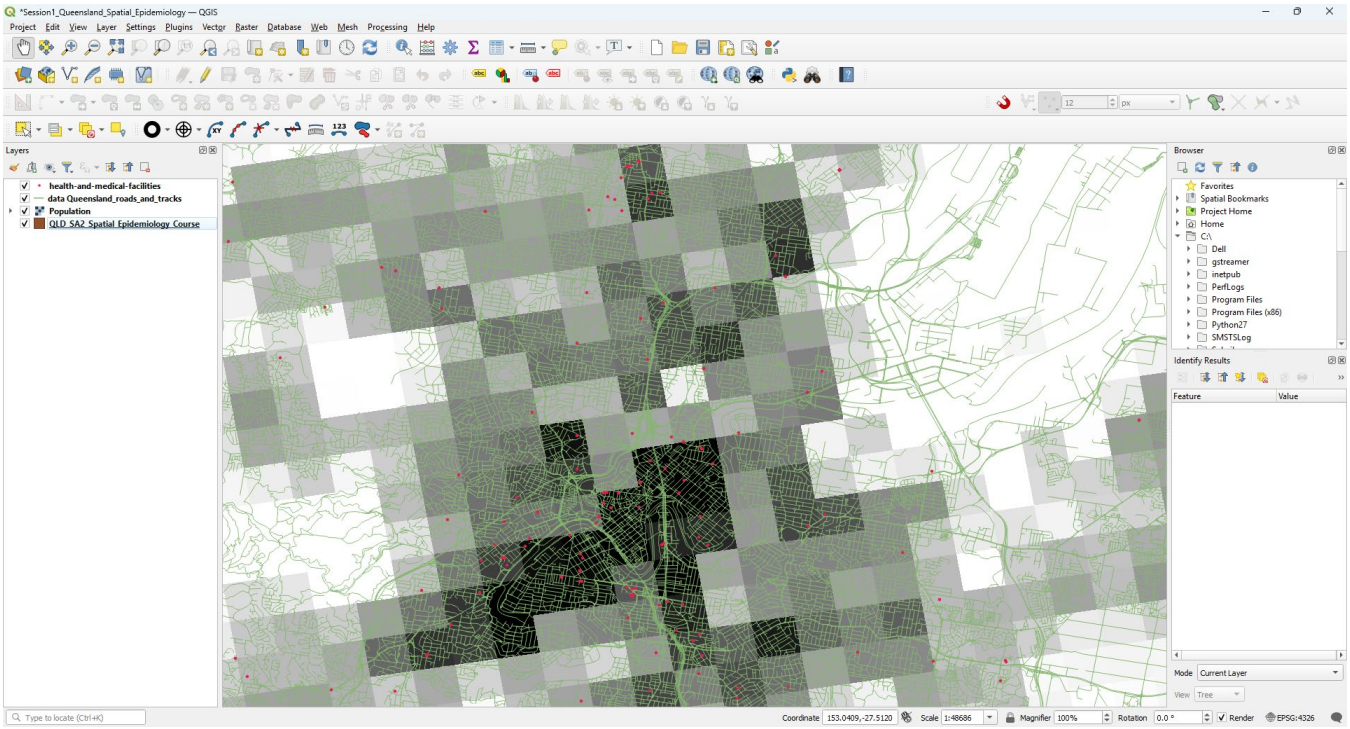
Step 2. Click the browse button beside Raster dataset(s).

Step 3. Navigate to the population raster supplied for the course.

Step 4. Select the file and click Add.

Step 5. Click Close.

Expected result: A gridded population surface should appear in the map and a new raster layer should appear in the Layers panel.



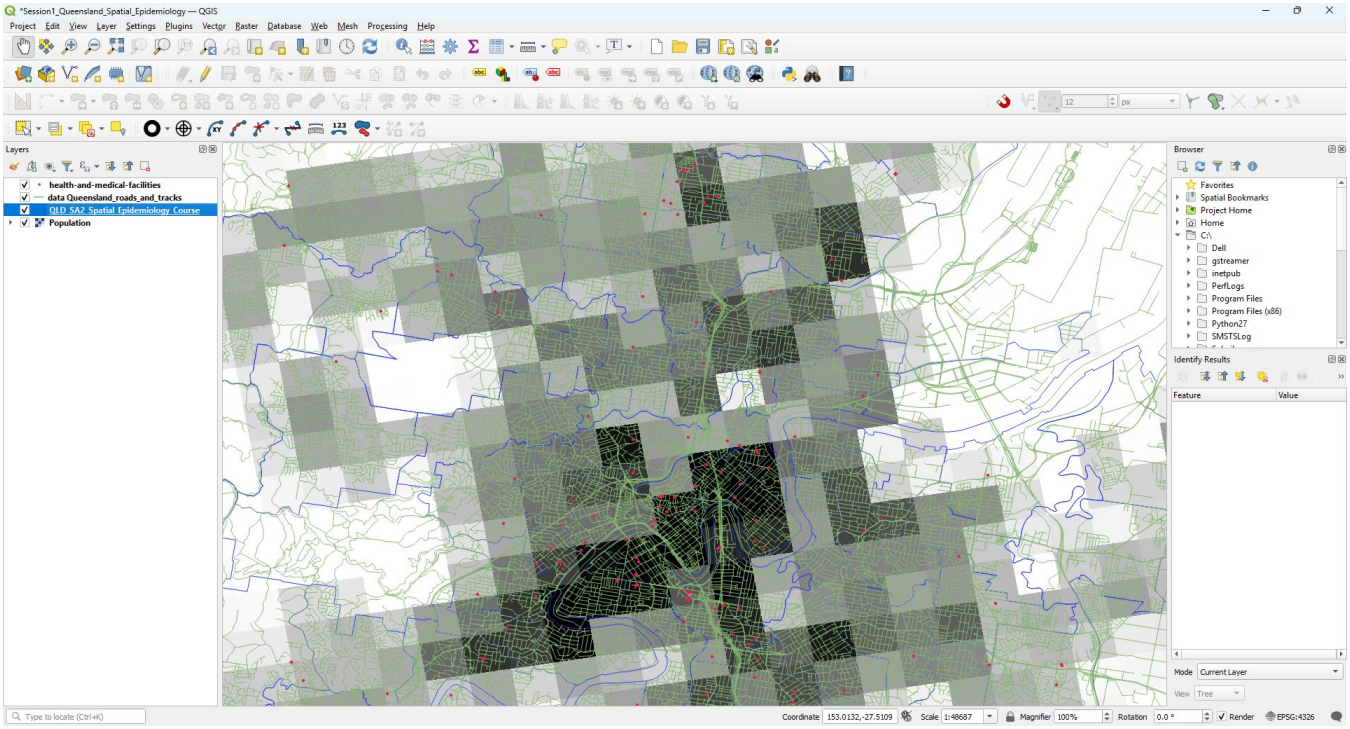
The population raster displayed in QGIS with the SA2, roads and health facility layers also listed.

8.2 Make the raster visible

Step 1. If the SA2 polygons hide the raster, temporarily untick the SA2 layer.

Step 2. Alternatively, open SA2 Layer Properties > Symbology and reduce the fill opacity or use a transparent fill.

Step 3. Keep the SA2 boundaries visible if possible so you can compare the raster with the administrative areas.



Population raster visible beneath transparent or outline-only SA2 polygons.

8.3 Zoom into raster cells

Step 1. Zoom into a populated area.

Step 2. Continue zooming until the raster structure becomes apparent.

Step 3. Use Identify Features on the raster and click a cell if the raster values are suitable for identification.

Each raster cell represents a small area. Its value describes population for that cell according to the supplied population product.

Key concept: An SA2 population variable gives one value for the whole polygon. A population raster shows how population is distributed within that polygon.

8.4 Why this matters in spatial epidemiology

Imagine a very large remote SA2. Most residents may live in one small town while the rest of the SA2 is almost uninhabited. If we treat the population as evenly distributed across the whole polygon, we may badly represent where people actually live.

Gridded population data can therefore be useful for accessibility analysis, exposure assessment, disease-rate denominators and other analyses where the within-area distribution of people matters.

Think: Look at one geographically large SA2. Does the population raster suggest that people are evenly distributed across it?

Part 9. Create your first epidemiological choropleth map

9.1 Map the simulated cases

Step 1. Turn the Queensland SA2 layer on.

Step 2. Right-click the SA2 layer and choose Properties.

Step 3. Select Symbology.

Step 4. Change the renderer from Single Symbol to Graduated.

Step 5. For Value, choose the field containing the cases.

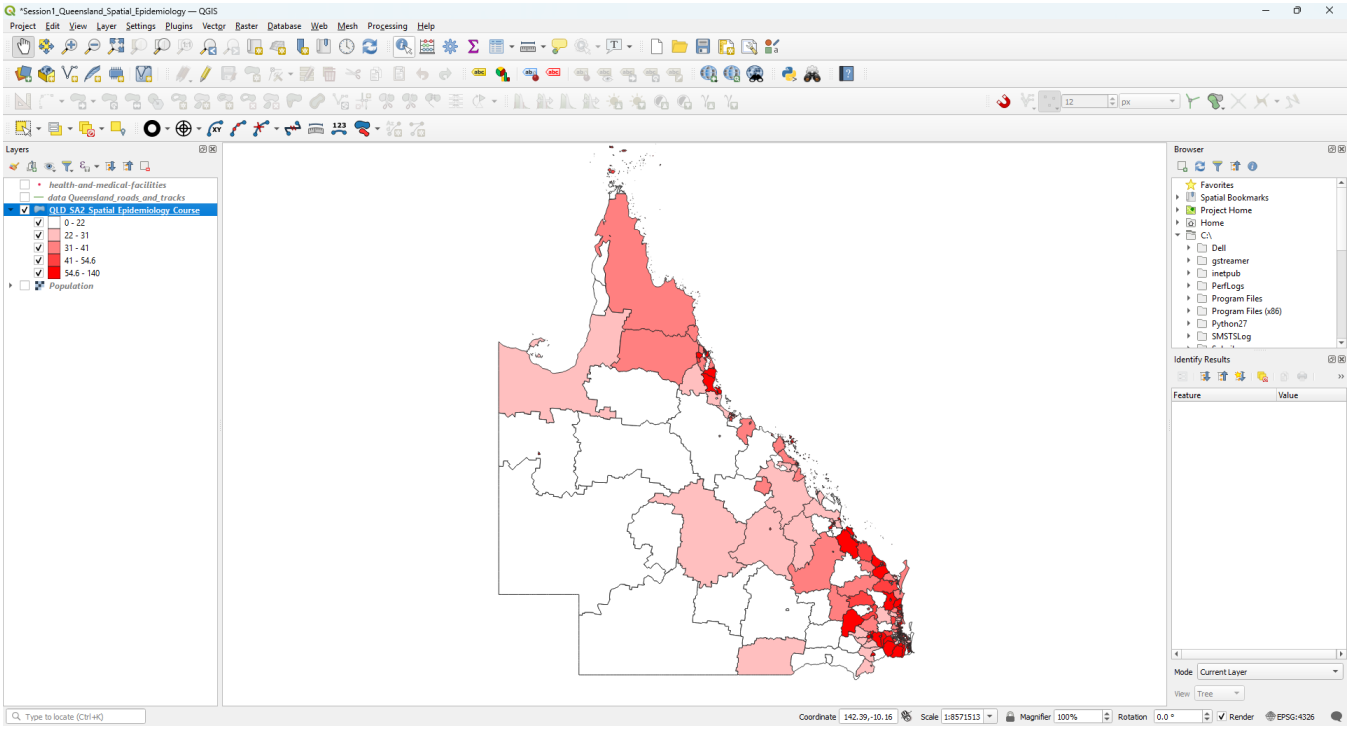
Step 6. Choose approximately five classes.

Step 7. Click Classify.

Step 8. Click Apply.

Step 9. Click OK.

Expected result: The SA2 polygons should now be shaded according to the value of the cases.



The graduated choropleth map of the cases across Queensland.

9.2 Interpret carefully

Look at the map before doing anything else.

Pause and interpret: Where are relatively high values located? Are there areas where neighbouring SA2s appear to have similar values? Are there isolated high values?

A choropleth map is an exploratory tool. It can help us see possible spatial patterns, but visual patterns alone do not tell us whether clustering is statistically significant.

Important: Do not interpret a visually striking map as proof of a causal relationship or a statistically significant cluster.

Part 10. Classification changes what the map looks like

Step 1. Open SA2 Layer Properties > Symbology again.

Step 2. Look at the Mode or classification method used for the graduated map.

Step 3. Try another classification method available in QGIS.

Step 4. Click Classify and Apply.

Step 5. Compare the new map with the previous map.

The underlying data have not changed, but the visual impression may change because the class boundaries have changed.

Key concept: Map design decisions influence interpretation. Always know how your values have been classified.

Part 11. Explore the explanatory variables

11.1 Map population

Step 1. Open SA2 Layer Properties > Symbology.

Step 2. Keep Graduated as the renderer.

Step 3. Choose the SA2 population field.

Step 4. Click Classify, Apply and OK.

Think: Compare the population choropleth with the population raster. What information is lost when the population is represented as one value for an entire SA2?

11.2 Map socioeconomic conditions

Step 1. Open Symbology again.

Step 2. Choose the SEIFA/IRSAD score field supplied in the course dataset.

Step 3. Classify the values and apply the map.

Pause and interpret: Do areas with similar socioeconomic values appear geographically concentrated? How might socioeconomic context be relevant to health outcomes?

11.3 Map remoteness

Remoteness is usually a category rather than a continuous measurement. Therefore, we will use Categorized rather than Graduated symbology.

Step 1. Open SA2 Layer Properties > Symbology.

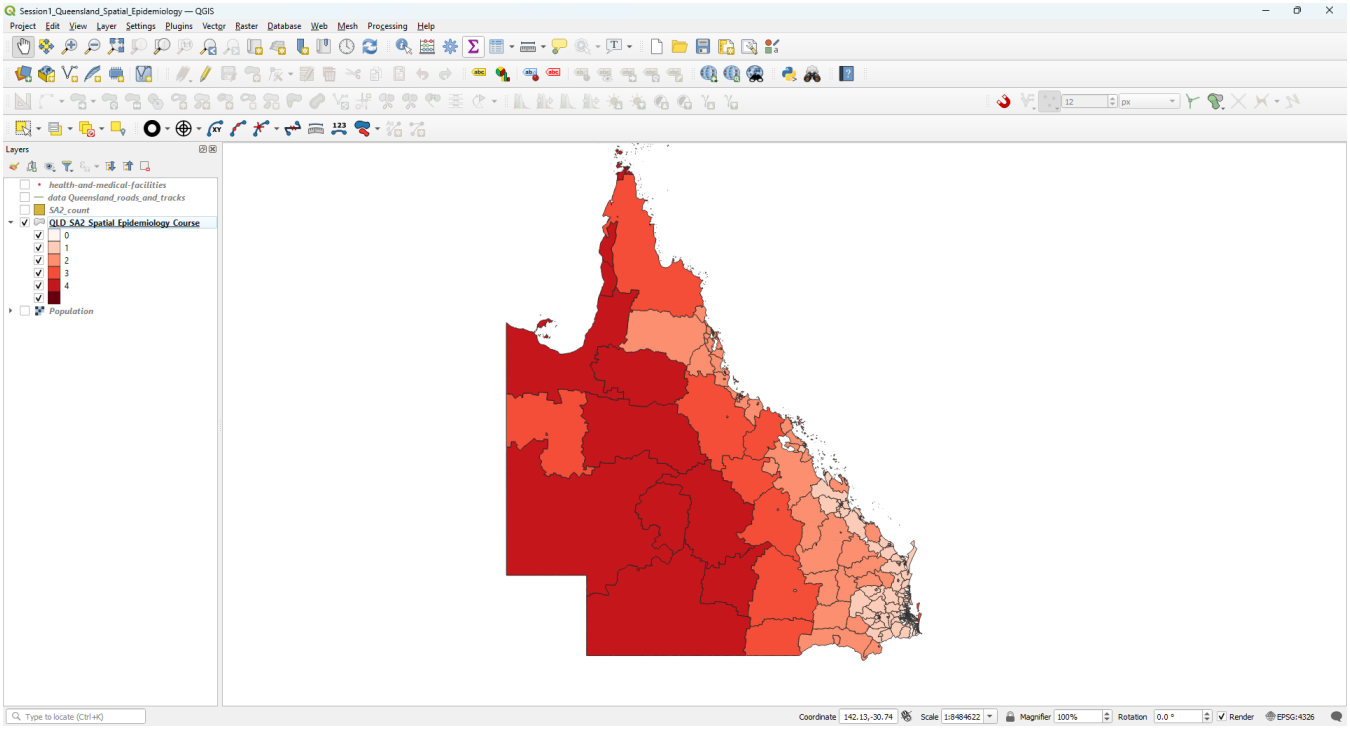
Step 2. Change the renderer to Categorized.

Step 3. Choose the remoteness field.

Step 4. Click Classify.

Step 5. Click Apply, then OK.

Expected result: SA2s should be grouped according to their remoteness category.



Queensland SA2s mapped by remoteness.

11.4 Map rainfall

Step 1. Open Symbology.

Step 2. Change back to Graduated if necessary.

Step 3. Choose the rainfall variable.

Step 4. Classify and apply the map.

Think: Rainfall is an environmental exposure that varies spatially. For which infectious diseases might rainfall plausibly be epidemiologically relevant? What other temporal information would you need before testing such a relationship?

Part 12. Select features using attributes

12.1 Select manually

Step 1. Choose the Select Features tool.

Step 2. Click an SA2.

Step 3. Hold the appropriate modifier key if you want to select several SA2s.

Step 4. Open the attribute table and display selected features.

Step 5. Clear the selection when finished.

12.2 Select using an expression

Step 1. Open the SA2 attribute table.

Step 2. Click Select features using an expression.

Step 3. Choose one suitable numeric variable such as population.

Step 4. Build a simple expression that selects areas above a chosen value.

Step 5. Click Select Features.

Step 6. Return to the map and inspect the selected areas.

Key concept: Attribute queries ask questions about the information stored in the table. Spatial queries ask questions about geographic relationships.

Part 13. A first spatial query: which SA2s contain health facilities?

13.1 Select by location

Step 1. Make sure the SA2 and health facility layers are both visible.

Step 2. Open Vector > Research Tools > Select by Location, or search for Select by Location in the Processing Toolbox if the menu location differs in your QGIS version.

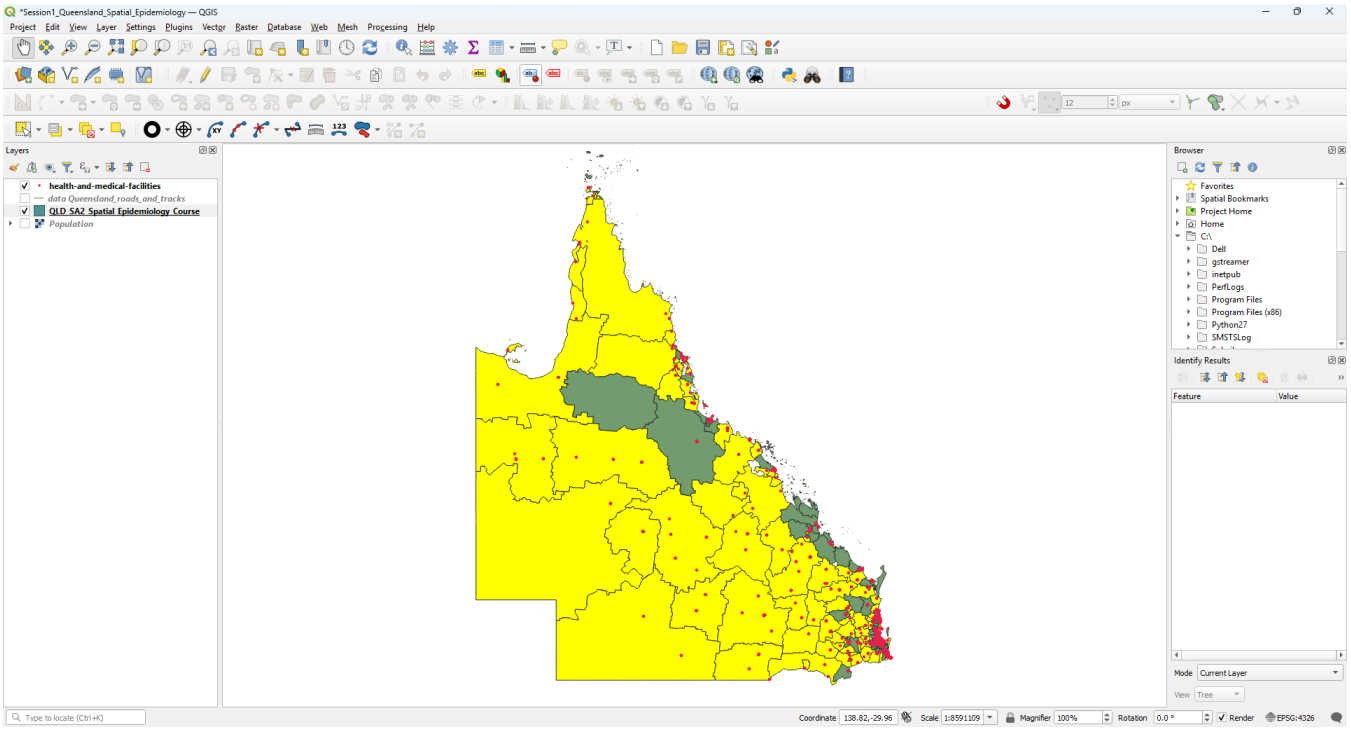
Step 3. For Select features from, choose the Queensland SA2 layer.

Step 4. Choose an appropriate geometric predicate such as contains or intersects.

Step 5. For By comparing to the features from, choose the health facilities layer.

Step 6. Run the tool.

Expected result: SA2s containing or intersecting at least one health facility should be selected.



Selected SA2 polygons after a Select by Location operation using health facilities.

Key concept: Spatial relationships allow datasets to be connected using location even when they do not share a common ID field.

Part 14. Optional if time allows: count health facilities in each SA2

Step 1. Open Processing > Toolbox.

Step 2. Search for Count points in polygon.

Step 3. Set the polygon layer to the Queensland SA2 layer.

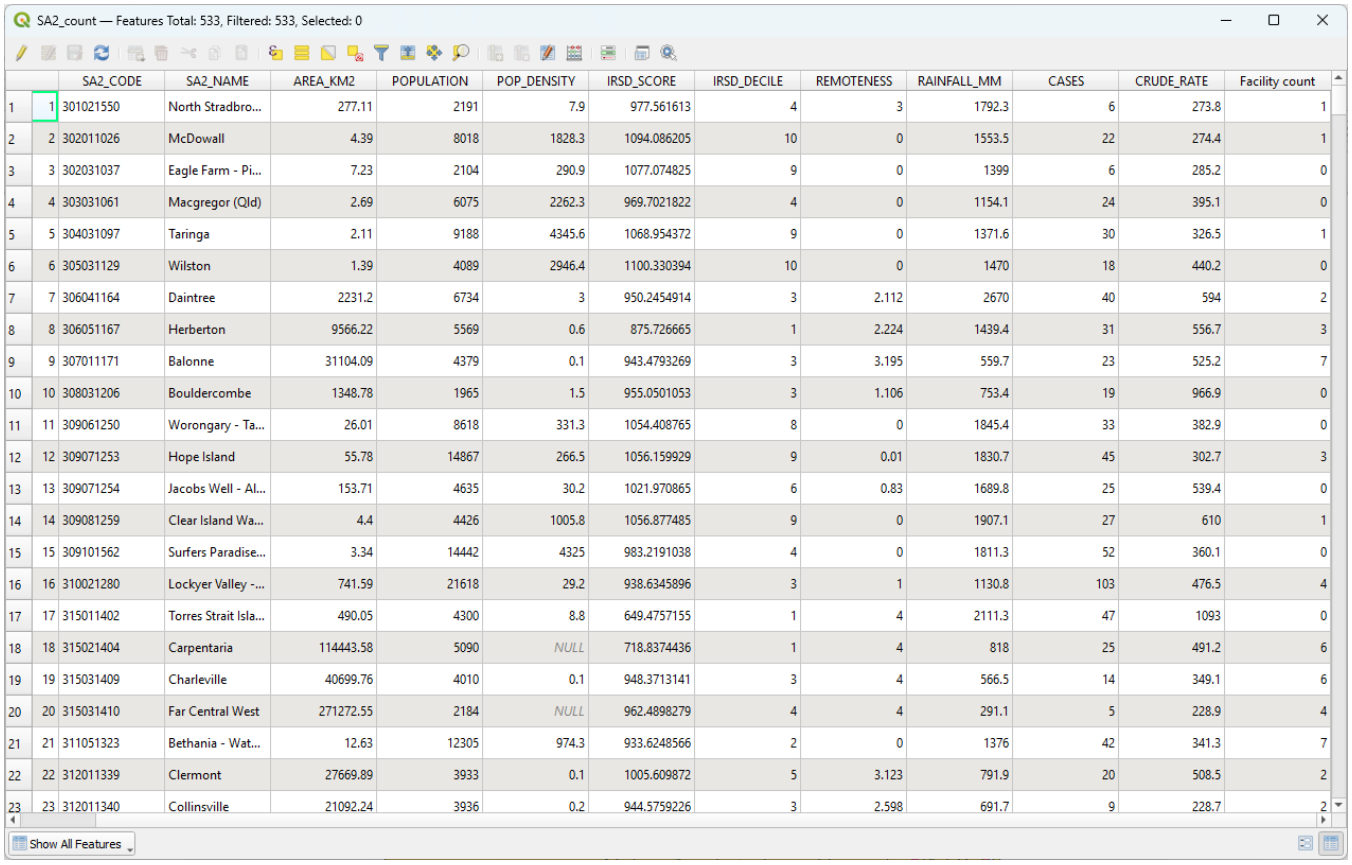
Step 4. Set the points layer to the health facilities layer.

Step 5. Leave the count field name at its default or give it a clear name such as facility_count.

Step 6. Run the tool.

Step 7. Open the output attribute table.

Expected result: A new polygon layer should contain a field giving the number of health facility points within each SA2.



		SA2_CODE	SA2_NAME	AREA_KM2	POPULATION	POP_DENSITY	IRSD_SCORE	IRSD_DECILE	REMOTENESS	RAINFALL_MM	CASES	CRUDE_RATE	Facility count
1	1	301021550	North Stradbro...	277.11	2191	7.9	977.561613	4	3	1792.3	6	273.8	1
2	2	302011026	McDowall	4.39	8018	1828.3	1094.086205	10	0	1553.5	22	274.4	1
3	3	302031037	Eagle Farm - Pi...	7.23	2104	290.9	1077.074825	9	0	1399	6	285.2	0
4	4	303031061	Macgregor (Qld)	2.69	6075	2262.3	969.7021822	4	0	1154.1	24	395.1	0
5	5	304031097	Taringa	2.11	9188	4345.6	1068.954372	9	0	1371.6	30	326.5	1
6	6	305031129	Wilston	1.39	4089	2946.4	1100.330394	10	0	1470	18	440.2	0
7	7	306041164	Daintree	2231.2	6734	3	950.2454914	3	2.112	2670	40	594	2
8	8	306051167	Herberton	9566.22	5569	0.6	875.726665	1	2.224	1439.4	31	556.7	3
9	9	307011171	Balonne	31104.09	4379	0.1	943.4793269	3	3.195	559.7	23	525.2	7
10	10	308031206	Bouldercombe	1348.78	1965	1.5	955.0501053	3	1.106	753.4	19	966.9	0
11	11	309061250	Worongary - Ta...	26.01	8618	331.3	1054.408765	8	0	1845.4	33	382.9	0
12	12	309071253	Hope Island	55.78	14867	266.5	1056.159929	9	0.01	1830.7	45	302.7	3
13	13	309071254	Jacobs Well - Al...	153.71	4635	30.2	1021.970865	6	0.83	1689.8	25	539.4	0
14	14	309081259	Clear Island Wa...	4.4	4426	1005.8	1056.877485	9	0	1907.1	27	610	1
15	15	309101562	Surfers Paradise...	3.34	14442	4325	983.2191038	4	0	1811.3	52	360.1	0
16	16	310021280	Lockyer Valley -...	741.59	21618	29.2	938.6345896	3	1	1130.8	103	476.5	4
17	17	315011402	Torres Strait Isla...	490.05	4300	8.8	649.4757155	1	4	2111.3	47	1093	0
18	18	315021404	Carpentaria	114443.58	5090	NULL	718.8374436	1	4	818	25	491.2	6
19	19	315031409	Charleville	40699.76	4010	0.1	948.3713141	3	4	566.5	14	349.1	6
20	20	315031410	Far Central West	271272.55	2184	NULL	962.4898279	4	4	291.1	5	228.9	4
21	21	311051323	Bethania - Wat...	12.63	12305	974.3	933.6248566	2	0	1376	42	341.3	7
22	22	312011339	Clermont	27669.89	3933	0.1	1005.609872	5	3.123	791.9	20	508.5	2
23	23	312011340	Collinsville	21092.24	3936	0.2	944.5759226	3	2.598	691.7	9	228.7	2

Count Points in Polygon dialog and/or resulting attribute table with the facility count field.

Pause and interpret: Is the number of facilities alone a good measure of healthcare accessibility? Consider SA2 size, population, facility capacity, service type and travel time.

Part 15. Optional if time allows: measure a distance

Step 1. Make sure the health facilities and roads are visible.

Step 2. Select the Measure Line tool.

Step 3. Click a starting location and then click a nearby health facility.

Step 4. Observe the distance and units reported by QGIS.

Think: Would this straight-line distance equal the actual distance a person travels by road? Why or why not?

Connection to later sessions: More advanced accessibility analysis can use road networks and travel time rather than simple straight-line distance.

Part 16. Create a simple map layout

16.1 Prepare the map

Return the SA2 symbology to the cases. Decide whether health facilities and roads help or distract from the final map. Keep only the layers needed for the message you want to communicate.

16.2 Create a print layout

Step 1. Click Project > New Print Layout.

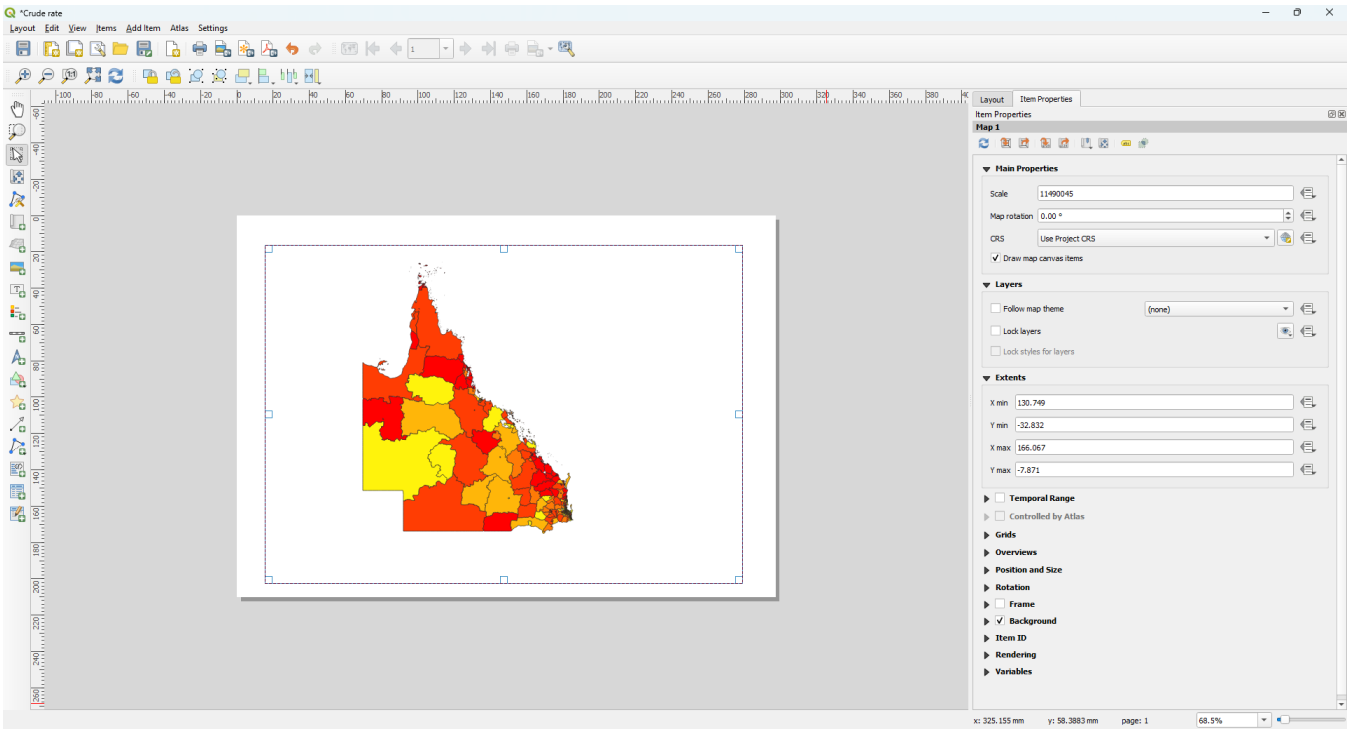
Step 2. Enter a name such as Cases_Map.

Step 3. Click OK.

Step 4. In the Layout window, choose Add Item > Add Map.

Step 5. Click and drag a rectangle on the page to create the map frame.

Expected result: Your QGIS map should appear inside the layout page.



QGIS Print Layout showing a map frame containing the Queensland disease crude rate

16.3 Add essential map elements

Step 1. Use Add Item > Add Label to create a clear map title.

Step 2. Use Add Item > Add Legend to add a legend.

Step 3. Use Add Item > Add Scale Bar to add a scale bar.

Step 4. If appropriate, add a north arrow using Add Picture and choose a north-arrow symbol.

Step 5. Move and resize the elements so the page is easy to read.

Good GIS habit: A map should communicate information, not simply demonstrate that you can add many map elements. Remove anything unnecessary.

16.4 Export your map

Step 1. In the Layout window, choose Layout > Export as PDF or Export as Image.

Step 2. Save the output in your Outputs folder.

Step 3. Return to the main QGIS window.

Step 4. Save your QGIS project again.

Part 17. Questions for discussion

Discuss the following questions with the person next to you or as a class:

- Why might a very large SA2 have a small population?
- What is the difference between the population value stored for an SA2 and the population raster?
- Why are health facilities represented as points and roads as lines?
- What is the difference between a layer CRS and the project CRS?
- Why can changing classification change the visual impression of a disease map?
- What can we learn from looking at a disease map that we cannot learn from an ordinary table?
- What can a map not tell us by itself?
- Why might straight-line distance to a hospital be misleading?
- How could roads, health facilities and population raster data be combined in an accessibility analysis?
- Why should an apparent spatial relationship between rainfall and disease not automatically be interpreted as causal?

Part 18. Common problems and troubleshooting

Problem	What to check
I added a layer but cannot see it.	Right-click the layer and choose Zoom to Layer. Check that the layer checkbox is ticked. Check whether another layer is covering it.
My health facility points are hidden.	Move the facility layer above polygon and raster layers. Increase the point marker size.
The population raster is hidden.	Turn off the SA2 fill temporarily, reduce polygon opacity, or move the raster above/below layers as appropriate.
My map suddenly looks different.	Check whether you changed symbology, classification, layer order or visibility.
The layers do not line up.	Check the CRS of each layer and the project CRS. Do not manually assign a new CRS just to force layers to align.
I cannot find a tool in the menu.	Open Processing > Toolbox and search for the tool by name. Menu locations can differ slightly between QGIS versions.
I closed the Layers or Browser panel.	Use View > Panels and tick the panel you want to restore.
QGIS says a layer is unavailable after reopening the project.	The source file may have been moved or renamed. Restore the original file location or repair the layer source.
I selected features and cannot remove the selection.	Use the Deselect Features from All Layers button or the relevant selection menu command.

Part 19. Session 1 recap

In this session you created a QGIS project and combined four important spatial data types:

- SA2 polygons – administrative areas with health and contextual attributes.
- Health facility points – discrete service locations.
- Road lines – transport-network features.
- Population raster – a gridded representation of where people live.

You also used attribute tables, Identify Features, layer symbology, a basic CRS inspection, graduated and categorized maps, attribute selection, a simple spatial query and a map layout.

Most important idea: Spatial epidemiology is not only about mapping disease. It is about linking health outcomes to people, place, environment and access in a way that respects geography.

Part 20. What you should have at the end

Before leaving the session, check that you have:

- A saved QGIS project: Session1_Queensland_Spatial_Epidemiology.qgz
- The Queensland SA2 layer loaded and explored.
- Health facilities loaded as points.
- Roads loaded as lines.
- Population loaded as a raster.
- At least one choropleth map of the simulated health outcome.
- Experience mapping SEIFA/IRSAD, remoteness or rainfall.
- A simple exported map in the Outputs folder.

Bridge to the next spatial epidemiology sessions

Session 1 focused on understanding and organising spatial data. Later sessions can build on this foundation to examine disease rates, spatial clustering, spatial autocorrelation, accessibility, spatial regression and other analytical methods.

When you begin those analyses, remember the questions introduced today: What does each spatial layer represent? At what geographic scale is it measured? How are people distributed within administrative areas? Which CRS is appropriate? And what can a map suggest versus what requires formal statistical analysis?